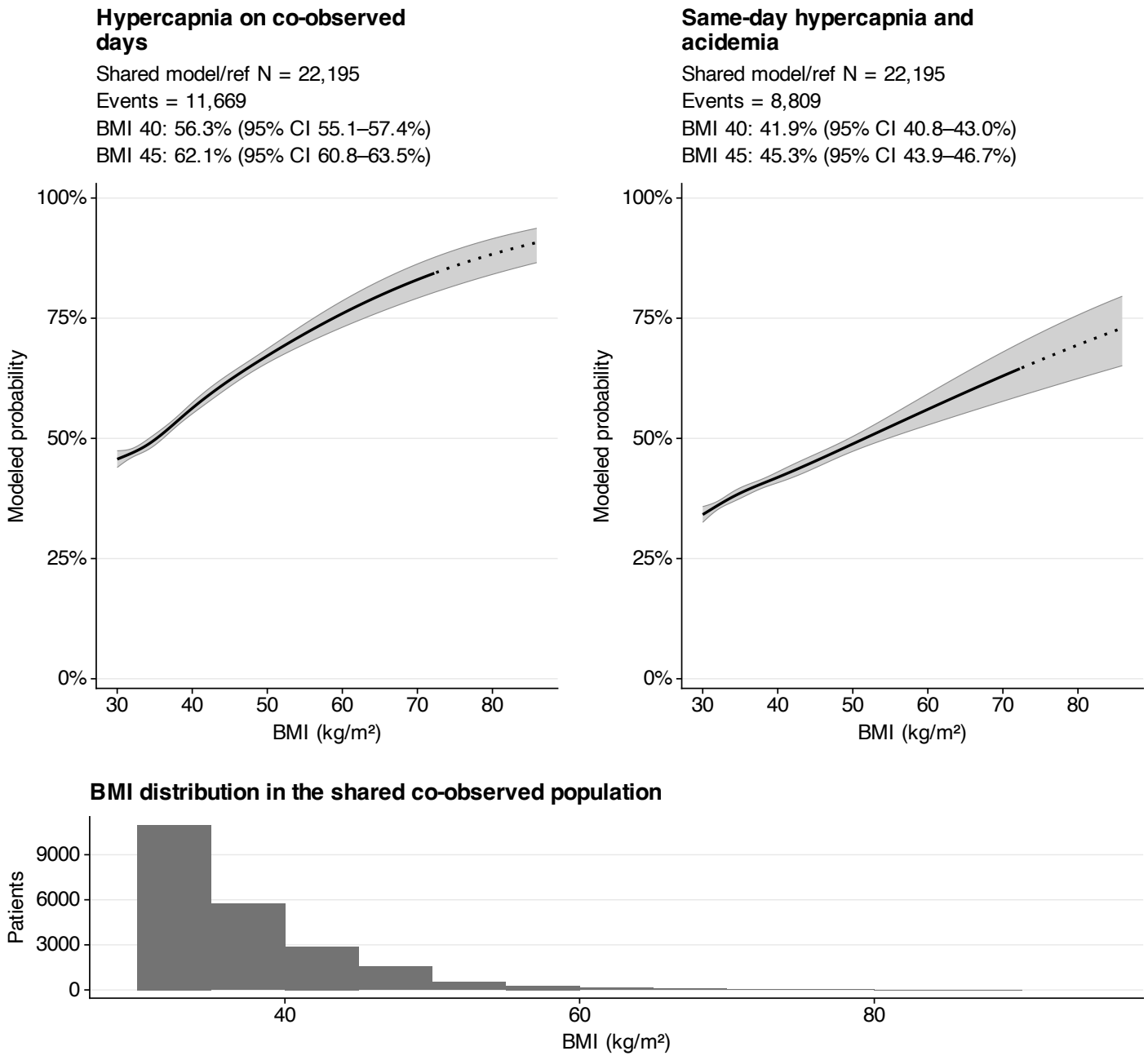
BMI and same-day arterial PCO_2/pH co-occurrence

Same recorded-day co-occurrence; not verified same-sample pairing



Supplemental Figure S1. Same recorded-day co-occurrence; not verified same-sample pairing. The primary source-entry population retains its fixed index encounters. Both endpoints use identical patients with valid arterial PCO_2 and pH on at least one recorded date, inherited adjustment and primary numeric BMI spline basis. Hypercapnia is $PaCO_2$ ≥45 mm Hg on a co-observed day; co-occurrence additionally requires pH <7.35 on that date, potentially from another sample. Different dates cannot be combined. Curves standardize to the shared outcome-evaluable population. Shading and printed intervals are pointwise model-based 95% intervals; dotted segments indicate local sparsity and dashed segments extrapolation. The lower panel counts patients by original BMI. Availability and unknown evidence are reported separately. No supplemental PIF is calculated; unavailable or failed estimates remain empty. Requiring both analytes adds selection. Findings may span different treatment periods and do not establish simultaneous acidemia, acute respiratory failure or compensation. Crossing curves require scientific review.